

Please check the examination details below before entering your candidate information

Candidate surname

mel@jstmaths.co.uk

Other names

**Pearson Edexcel
International GCSE**

Centre Number

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Candidate Number

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Tuesday 21 May 2019

Morning (Time: 2 hours)

Paper Reference **4MA1/1H**

Mathematics A
Level 1/2
Paper 1H
Higher Tier

WORKED
SOLUTIONS



You must have:

Ruler graduated in centimetres and millimetres, protractor, compasses, pen, HB pencil, eraser, calculator. Tracing paper may be used.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Without sufficient working, correct answers may be awarded no marks.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may be used.**
- You must **NOT** write anything on the formulae page.
Anything you write on the formulae page will gain NO credit.

Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

International GCSE Mathematics

Formulae sheet – Higher Tier

Arithmetic series

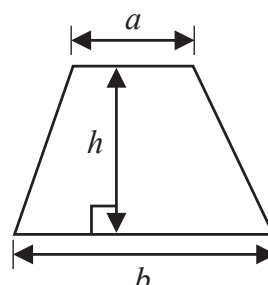
Sum to n terms, $S_n = \frac{n}{2} [2a + (n-1)d]$

The quadratic equation

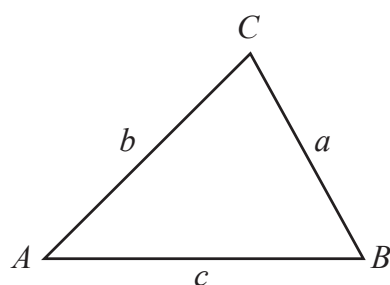
The solutions of $ax^2 + bx + c = 0$ where $a \neq 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Area of trapezium = $\frac{1}{2}(a+b)h$



Trigonometry



In any triangle ABC

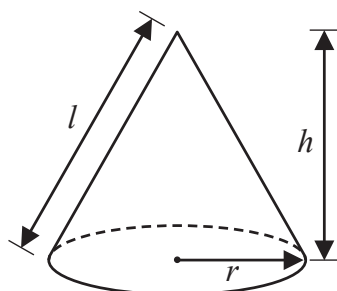
Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Cosine Rule $a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

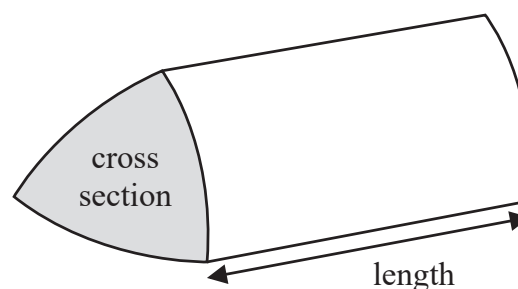
Volume of cone = $\frac{1}{3}\pi r^2 h$

Curved surface area of cone = $\pi r l$



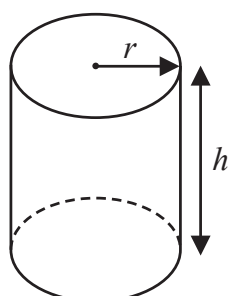
Volume of prism

= area of cross section $\times$ length



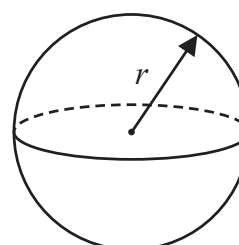
Volume of cylinder = $\pi r^2 h$

Curved surface area of cylinder = $2\pi r h$



Volume of sphere = $\frac{4}{3}\pi r^3$

Surface area of sphere = $4\pi r^2$



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Answer ALL TWENTY FOUR questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Show that $4\frac{2}{3} \div 1\frac{1}{9} = 4\frac{1}{5}$

$$4\frac{2}{3} = \frac{14}{3} \quad 1\frac{1}{9} = \frac{10}{9}$$

$$\frac{14}{3} \div \frac{10}{9} = \frac{14}{3} \times \frac{9}{10}$$

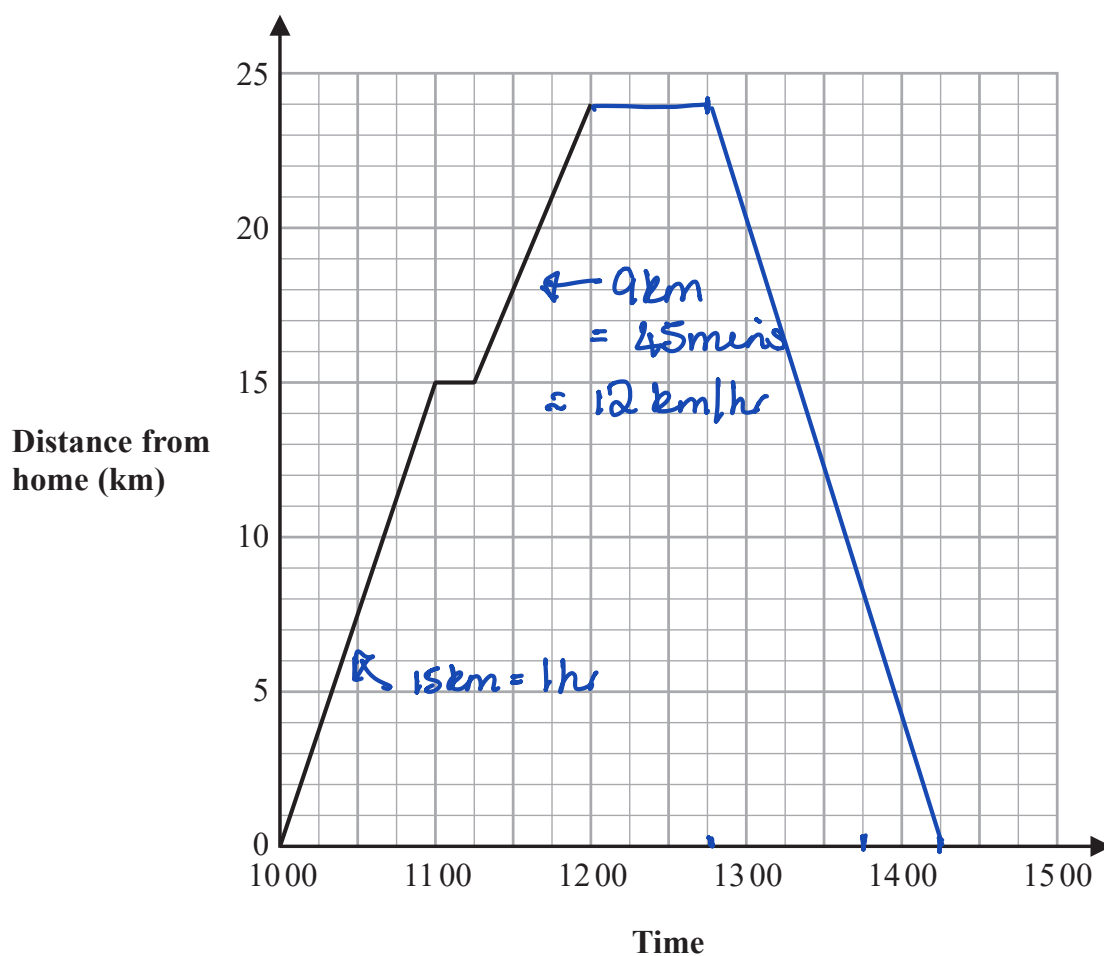
$$= \frac{21}{5}$$

$$\frac{21}{5} = 4\frac{1}{5} \text{ as required.}$$

(Total for Question 1 is 3 marks)



- 2 Jalina left her home at 1000 to cycle to a park.
On her way to the park, she stopped at a friend's house and then continued her journey to the park.
Here is the distance-time graph for her journey to the park.



- (a) On her journey to the park, did Jalina cycle at a faster speed before or after she stopped at her friend's house?
Give a reason for your answer.

Before, the gradient of the line before she stopped is steeper. (15 km/hr > 12 km/hr)

(1)

Jalina stayed at the park for 45 minutes.

She then cycled, without stopping, at a constant speed of 16 km/h from the park back to her home.

- (b) Show all this information on the distance-time graph.

24 km distance to home

$$16 \text{ km} = 1 \text{ hr}$$

$$8 \text{ km} = 30 \text{ mins}$$

$$24 \text{ km} = 1\frac{1}{2} \text{ hours}$$

(2)

- (c) Work out Jalina's average cycling speed, in kilometres per hour, for the complete journey to the park and back.

Do **not** include the times when she was not cycling in your calculation.

Give your answer correct to 1 decimal place.

$$\text{Total distance} = 24 + 24 = 48 \text{ km}$$

$$\text{Total time} = 4 \text{ hours } 15 \text{ mins}$$

$$\text{less stopping time } 1 \text{ hour} = 3 \text{ hr } 15 \text{ mins}$$

$$3.25 \text{ hr}$$

$$\text{Speed} = 48 \div 3.25$$

$$= 14.769...$$

↑

(1 dp)

$$\underline{\quad 14.8 \quad} \text{ km/h}$$

(3)

(Total for Question 2 is 6 marks)



3 (a) Simplify $e^9 \div e^5$

$$e^{9-5}$$

$$e^4$$

(1)

(b) Simplify $(y^2)^8$

$$y^{2 \times 8}$$

$$y^{16}$$

(1)

(c) Expand and simplify $(x + 9)(x - 2)$

$$x \times x = x^2$$

$$x \times -2 = -2x$$

$$9 \times x = 9x$$

$$9 \times -2 = -18$$

$$x^2 + 7x - 18$$

(2)

(d) Factorise fully $16c^4p^2 + 20cp^3$

$$4cp^2(4c^3 + 5p)$$

(2)

(Total for Question 3 is 6 marks)



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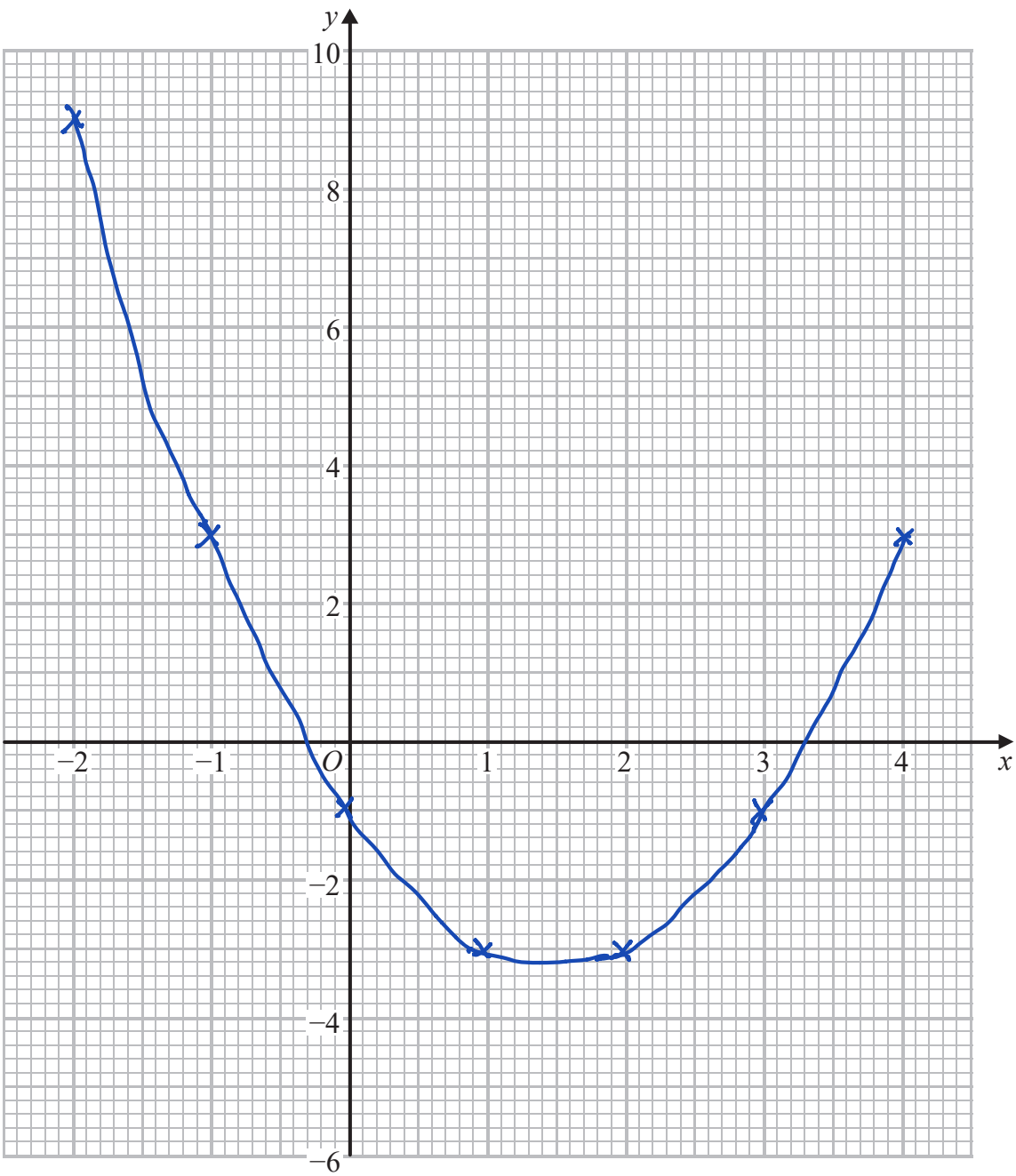
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4 (a) Complete the table of values for $y = x^2 - 3x - 1$

x	-2	-1	0	1	2	3	4
y	9	3	-1	-3	-3	-1	3

(2)

(b) On the grid, draw the graph of $y = x^2 - 3x - 1$ for all values of x from -2 to 4



(2)

(Total for Question 4 is 4 marks)



- 5 Becky has a biased 6-sided dice.

The table gives information about the probability that, when the dice is thrown, it will land on each number.

Number	1	2	3	4	5	6
Probability	$2x$	0.18	$2x$	$3x$	0.26	x

Becky is going to throw the dice 200 times.

Work out an estimate for the number of times that the dice will land on an even number.

$$0.18 + 0.26 = 0.44$$

$$1 - 0.44 = 0.56$$

$$8x = 0.56$$

$$x = 0.07$$

$$\begin{aligned} P(\text{Even}) &= 0.18 + 3 \times 0.07 + 0.07 \\ &= 0.18 + 0.21 + 0.07 \\ &= 0.46 \end{aligned}$$

$$200 \times 0.46 = 92$$

92

(Total for Question 5 is 4 marks)



- 6 The diagram shows a solid cuboid made from wood.

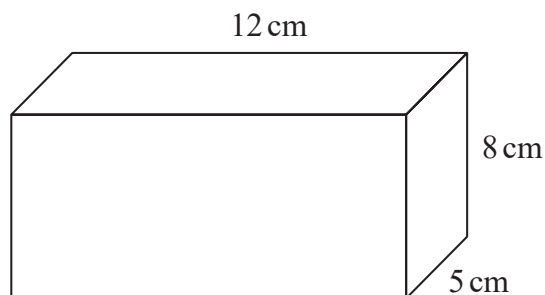


Diagram **NOT**
accurately drawn

The wood has density 0.7 g/cm^3

Work out the mass of the cuboid.

$$\text{Volume} = 5 \times 8 \times 12 = 480 \text{ cm}^3$$

$$D = \frac{m}{V} \quad \text{so} \quad m = D \times V$$

$$m = 0.7 \times 480 \\ = 336$$

..... **336** grams

(Total for Question 6 is 3 marks)



- 7 (a) Write 5.7×10^6 as an ordinary number.

5700000

(1)

- (b) Write 0.004 in standard form.

4×10^{-4}

(1)

- (c) Work out $\frac{2 \times 10^4 + 3 \times 10^5}{6.4 \times 10^{-2}}$

$\frac{320000}{0.064}$

= 5000 000

5×10^6

(2)

(Total for Question 7 is 4 marks)

- 8 On 1st January 2016 Li bought a boat for \$170 000
The value of the boat depreciates by 8% per year.

0.92

Work out the value of the boat on 1st January 2019
Give your answer correct to the nearest dollar.

2016 → 2017 → 2018 → 2019 = 3 years

170000×0.92^3

= 132376.96

↑
nearest \$

\$ 132377

(Total for Question 8 is 3 marks)



- 9 The diagram shows a shape made from a right-angled triangle and a semicircle.

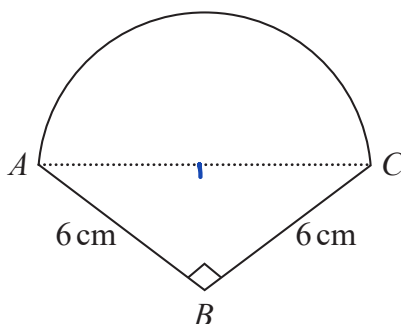


Diagram **NOT**
accurately drawn

AC is the diameter of the semicircle.

$BA = BC = 6 \text{ cm}$

Angle $ABC = 90^\circ$

Work out the area of the shape.

Give your answer correct to 1 decimal place.

$$AC^2 = 6^2 + 6^2 \quad AC = \sqrt{72} \\ = 8.48 \dots$$

$$\text{so Diameter} = 8.48 \dots \quad \text{radius} = 4.2426 \dots$$

$$\begin{aligned} &\text{Area of triangle} + \text{area of semicircle} \\ &= \frac{1}{2} 6 \times 6 + \frac{1}{2} \pi \times 4.24 \dots^2 \end{aligned}$$

$$= 18 + 28.27 \dots$$

$$= 46.274 \dots$$

↑

(1dp)

$$46.3 \dots \text{ cm}^2$$

(Total for Question 9 is 5 marks)



10 $A = 2^n \times 3 \times 5^m$

Write $8A$ as a product of powers of its prime factors.

$$8 = 2^3$$

$$\begin{aligned} \text{so } 8A &= 2^3 \times 2^n \times 3 \times 5^m \\ &= 2^{3+n} \times 3 \times 5^m \end{aligned}$$

$$2^{3+n} \times 3 \times 5^m$$

(Total for Question 10 is 2 marks)

11 $C = b - a$

$a = 6$ correct to the nearest integer

$b = 15$ correct to the nearest 5

Work out the upper bound for the value of C

Show your working clearly.

$$\begin{array}{ccc} & a = 6 & \\ \swarrow & & \searrow \\ 5.5 & & 6.5 \end{array}$$

$$\begin{array}{ccc} & b = 15 & \\ \swarrow & & \searrow \\ 12.5 & & 17.5 \end{array}$$

$$\begin{aligned} C &= 17.5 - 5.5 \\ &= 12 \end{aligned}$$

$$12$$

(Total for Question 11 is 3 marks)



12 (a) Factorise $2x^2 - 7x + 6$

$$2 \times 6 = 12$$

1, 12
2, 6
3, 4

$$2x^2 - 4x - 3x + 6$$

$$= 2x(x - 2) - 3(x - 2)$$

$$= (2x - 3)(x - 2)$$

$$(2x - 3)(x - 2)$$

(2)

(b) Solve $\frac{4m + 9}{3} = 7 - 2m$

Show clear algebraic working.

$$4m + 9 = 3(7 - 2m)$$

$$4m + 9 = 21 - 6m$$

$$10m = 12$$

$$m = \frac{12}{10}$$

$$m = 1.2$$

(4)

(c) Write $\frac{\sqrt[4]{y}}{y}$ in the form y^b where b is a fraction.

$$\sqrt[4]{y} = y^{1/4}$$

$$y^{1/4 - 1} = y^{-3/4}$$

$$y^{-3/4}$$

(2)

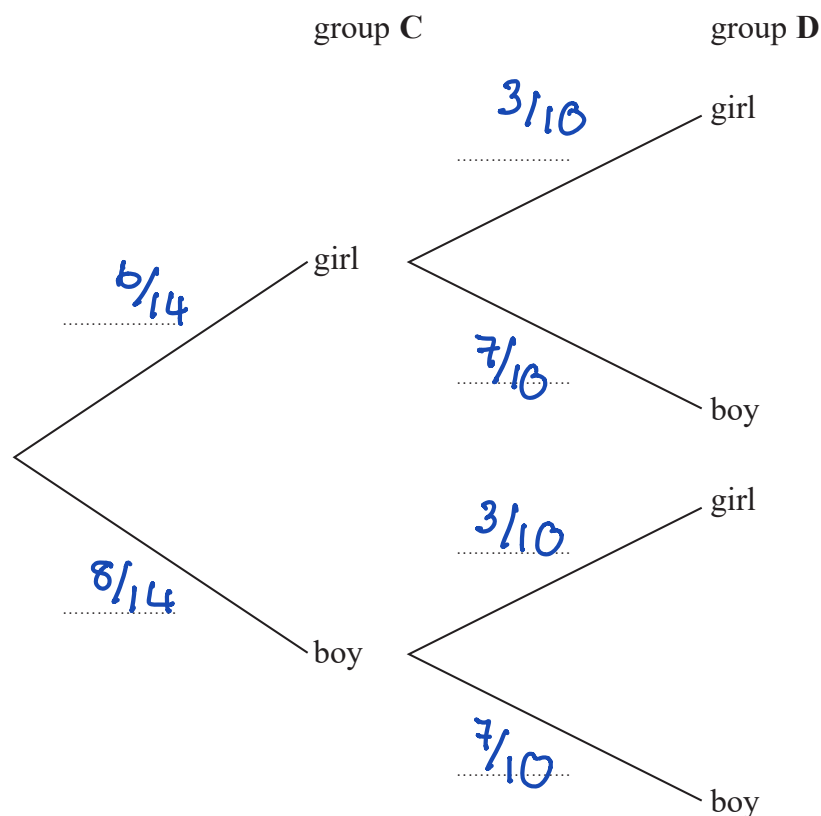
(Total for Question 12 is 8 marks)



- 13 In group C, there are 6 girls and 8 boys. 14
 In group D, there are 3 girls and 7 boys. 10

A team is made by picking at random one child from group C and one child from group D.

- (a) Complete the probability tree diagram.



(2)

- (b) Work out the probability that there are two boys in the team.

$$P(B, B) = \frac{8}{14} \times \frac{7}{10}$$

$$= \frac{4}{10}$$

$$\frac{4}{10} = \frac{2}{5}$$

(2)



After the first team has been picked, a second team is picked.

One child is picked at random from the children left in group C and one child is picked at random from the children left in group D.

(c) Work out the probability that there are two boys in each of the two teams.

$$\frac{7}{13} \times \frac{6}{9} = \frac{42}{117}$$

$$\frac{42}{117} \times \frac{2}{5} = \frac{84}{585}$$

$$\frac{28}{195}$$

(3)

(Total for Question 13 is 7 marks)

14 $\mathcal{E} = \{\text{positive integers less than } 20\}$

$$A = \{x : x < 12\} \quad 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ 10 \ 11$$

$$B = \{x : 7 \leq x < 16\} \quad 7 \ 8 \ 9 \ 10 \ 11 \ 12 \ 13 \ 14 \ 15$$

(a) List the members of $A \cap B$

$$7, 8, 9, 10, 11$$

(2)

C is a set such that $C \subset A$ and $n(C) = 3$

Given that all members of C are even numbers,

(b) list the members of one possible set C .

any three from

$$2, 4, 6, 8, 10$$

(1)

(Total for Question 14 is 3 marks)



- 15 Use algebra to show that the recurring decimal $0.2\dot{5}\dot{4} = \frac{14}{55}$

$$\begin{array}{r} 100x = 25.454\dots \\ x = 0.2545454\dots \\ \hline 99x = 25.2 \end{array}$$

$$x = \frac{252}{99}$$

$$\frac{25.2}{99} = \frac{252}{990}$$

$$\frac{252}{990} = \frac{126}{495}$$

$$\frac{126}{495} = \frac{14}{55} \text{ as required}$$

(Total for Question 15 is 2 marks)

- 16 Here are the first five terms of an arithmetic sequence.

$$7 \quad 10 \quad 13 \quad 16 \quad 19$$

$\xrightarrow{\quad 3 \quad}$
 $\xrightarrow{\quad 3 \quad}$

Find the sum of the first 100 terms of this sequence.

$$\text{Common difference} = 3 \quad a = 7$$

$$\frac{100}{2} (2 \times 7 + (100-1) \times 3)$$

$$= 50 (14 + 297)$$

$$= 50 \times 311$$

$$= 15550$$

$$15550$$

(Total for Question 16 is 2 marks)



17 A and B are two similar vases.

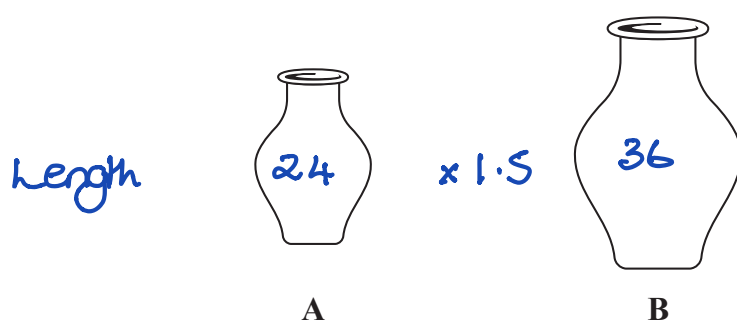


Diagram NOT
accurately drawn

Vase A has height 24 cm.

Vase B has height 36 cm.

Vase A has a surface area of 960 cm^2

(a) Work out the surface area of vase B.

$$\begin{aligned} \text{length SF} &= 1.5 \\ \text{area SF} &= 1.5^2 \\ &= 2.25 \end{aligned}$$

$$\text{Area B} = 960 \times 2.25$$

$$\underline{\underline{2160}} \text{ cm}^2$$

(2)

Vase B has a volume of $V \text{ cm}^3$

(b) Find in terms of V , an expression for the volume, in cm^3 , of vase A.

$$\text{Volume SF} = 1.5^3$$

$$\text{Vol B} = \text{Vol A} \times 1.5^3$$

$$\text{Vol A} = \frac{V}{1.5^3}$$

$$= \frac{8V}{27}$$

$$\left(V \div \frac{27}{8} = \frac{8V}{27} \right)$$

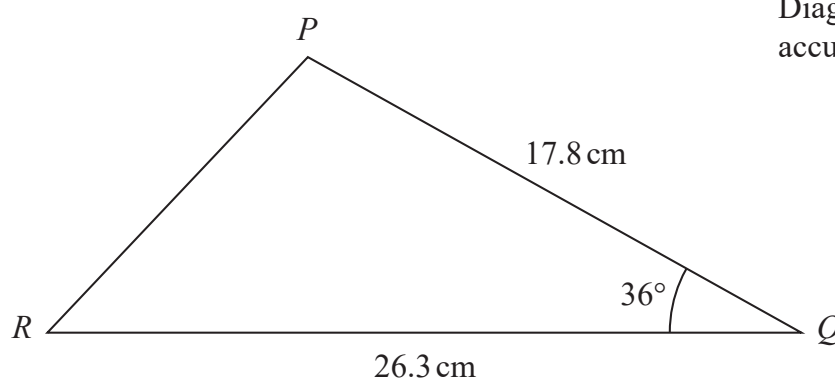
$$\underline{\underline{A = \frac{8V}{27}}} \text{ cm}^3$$

(2)

(Total for Question 17 is 4 marks)



18 The diagram shows triangle PQR .



Calculate the length of PR .

Give your answer correct to 3 significant figures.

$$PR^2 = 17.8^2 + 26.3^2 - 2 \times 17.8 \times 26.3 \times \cos 36$$
$$= 251.06 \dots$$

$$PR = \sqrt{251.06}$$
$$= 15.844 \dots$$

↑
(3sf)

15.8 cm

(Total for Question 18 is 3 marks)

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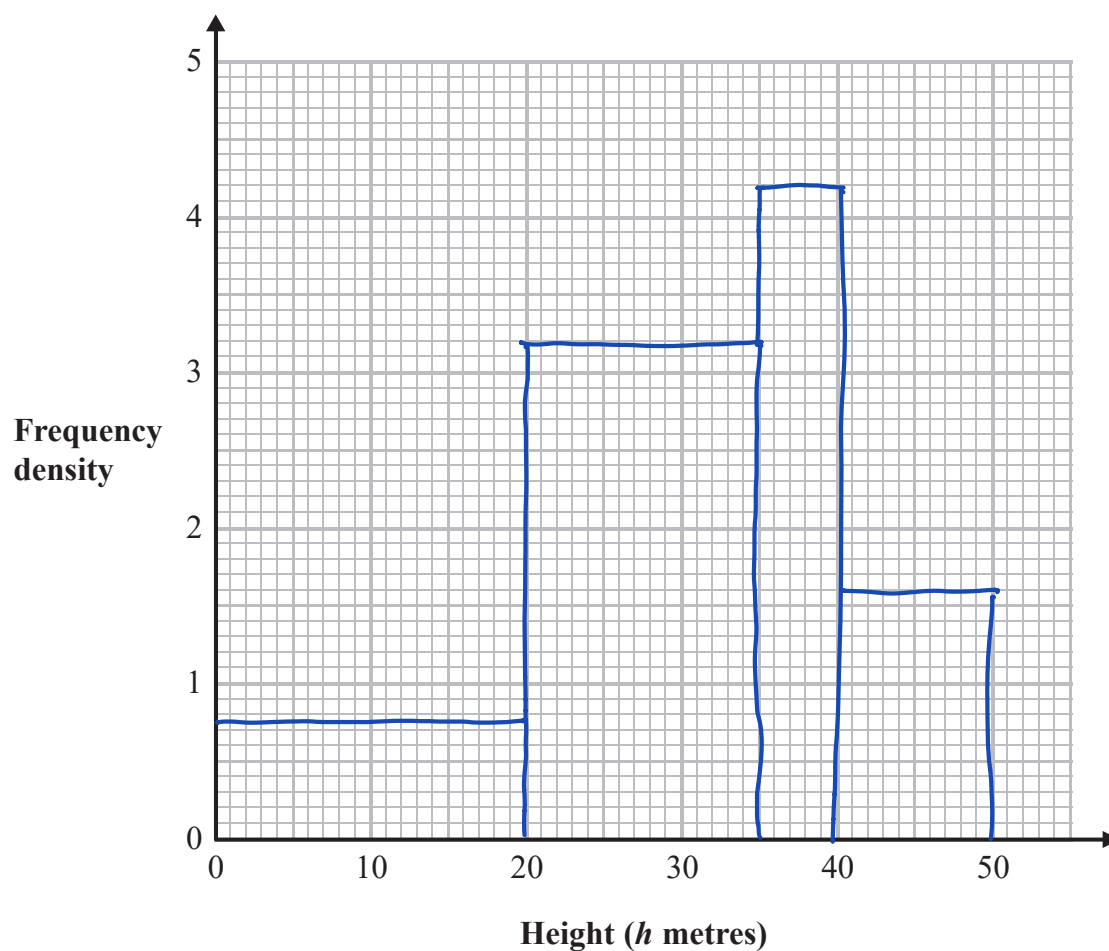
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19 The table gives information about the heights of some trees.

Freq. Density
 $15 \div 20 = 0.75$
 $48 \div 15 = 3.2$
 $21 \div 5 = 4.2$
 $16 \div 10 = 1.6$

Height (h metres)	Frequency
$0 < \overset{20}{h} \leq 20$	15
$20 < \overset{15}{h} \leq 35$	48
$35 < \overset{5}{h} \leq 40$	21
$40 < \overset{10}{h} \leq 50$	16

On the grid, draw a histogram for this information.



(Total for Question 19 is 3 marks)



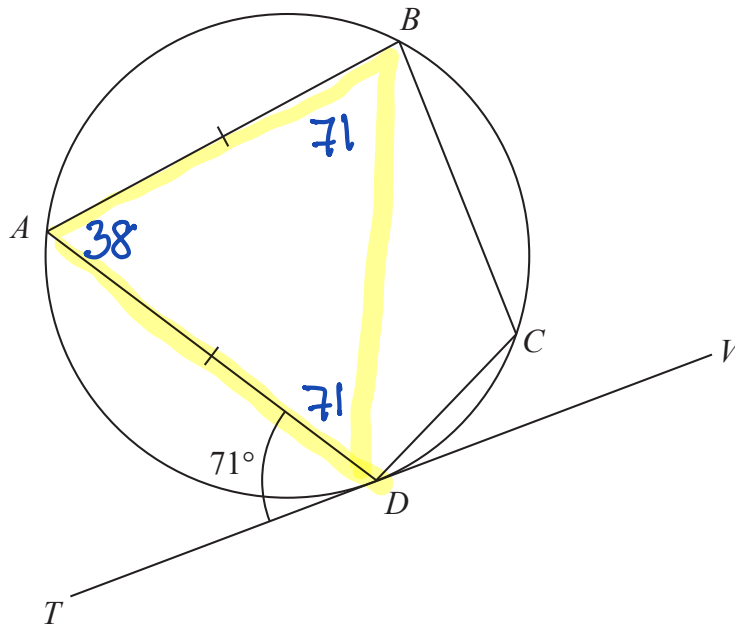


Diagram **NOT**
accurately drawn

A, B, C and D are points on a circle.
 TDV is the tangent to the circle at D .

$AB = AD$
Angle $ADT = 71^\circ$

Work out the size of angle BCD .
Give a reason for each stage of your working.

$\angle ABD = 71^\circ$ alternate segment theorem

$\triangle ABD$

$$\hat{A}BD = \hat{B}DA = 71$$

2 angles in an
isosceles are equal

$$\hat{B}AD = 180 - 142 = 38$$

angles in a triangle
= 180

$$\therefore \angle BCD = 180 - 38 = 142$$

opposite angles in a cyclic quadrilateral
add to 180

142

(Total for Question 20 is 5 marks)



- 21 A solid is made from a hemisphere and a cylinder.
The plane face of the hemisphere coincides with the upper plane face of the cylinder.

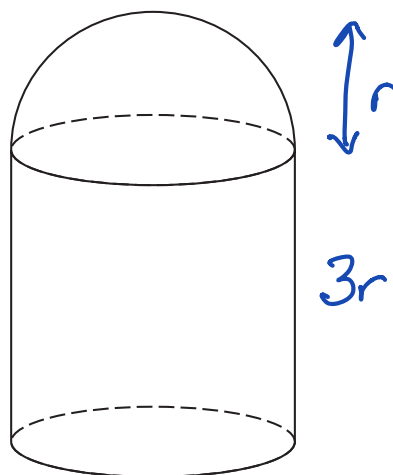


Diagram **NOT**
accurately drawn

$$\begin{aligned} r &: h \\ 1 &: 3 \\ \therefore h &= 3r \end{aligned}$$

The hemisphere and the cylinder have the same radius.

The ratio of the radius of the cylinder to the height of the cylinder is 1 : 3

Given that the solid has volume $792\pi \text{ cm}^3$
work out the height of the solid.

$$\frac{1}{2} \times \frac{4}{3} \pi r^3 + \pi r^2 \times 3r = 792\pi$$

$$\frac{4}{6} \pi r^3 + 3\pi r^3 = 792\pi$$

$$\frac{22}{6} r^3 = 792$$

$$r^3 = \frac{792 \times 6}{22} = 216$$

$$r = \sqrt[3]{216} = 6$$

$$\therefore h = 18$$

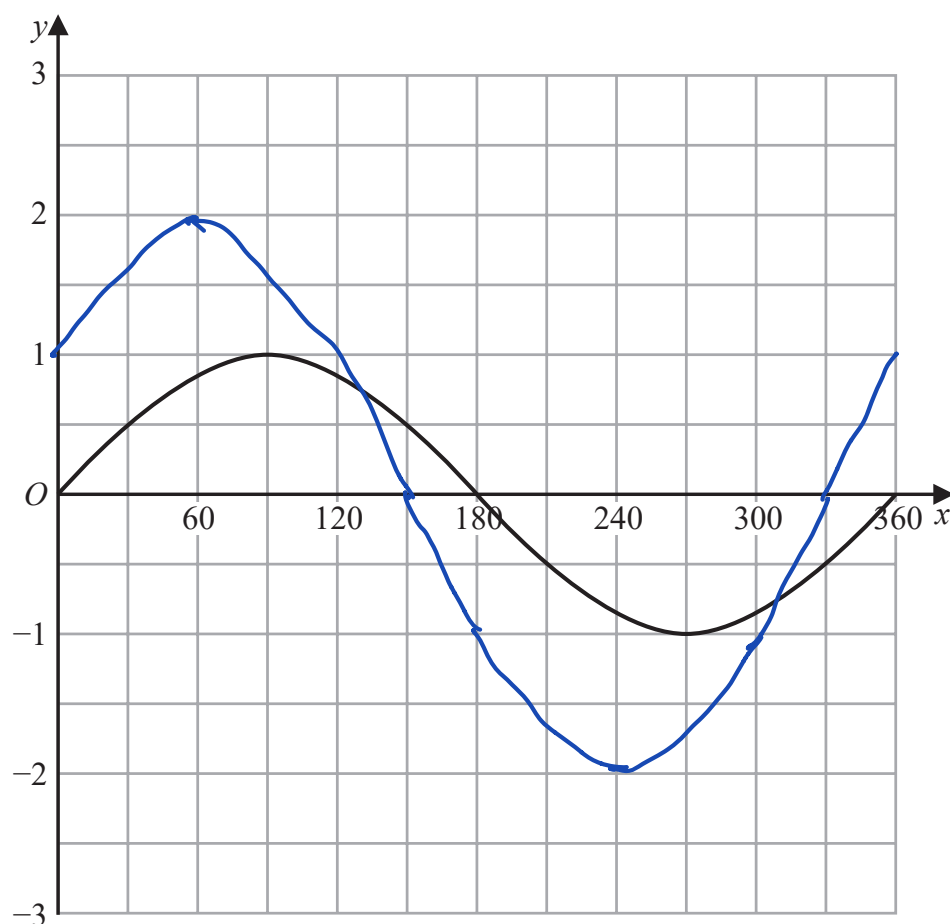
$$\text{Height of solid} = 18 + 6$$

$$\dots\dots\dots 24 \dots\dots\dots \text{cm}$$

(Total for Question 21 is 5 marks)



22 The graph of $y = \sin x^\circ$ for $0 \leq x \leq 360$ is drawn on the grid.



(a) On the grid, draw the graph of $y = 2\sin(x + 30)^\circ$ for $0 \leq x \leq 360$



(2)

(b) (i) Write $x^2 - 6x + 10$ in the form $(x - a)^2 + b$ where a and b are integers.

$$(x - 3)^2 - 9 + 10$$

$$(x - 3)^2 + 1$$

(2)

(ii) Hence, describe fully the single transformation that maps the curve with equation $y = x^2$ onto the curve with equation $y = x^2 - 6x + 10$

3 → 1 ↑ Translation of $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$

(2)

(Total for Question 22 is 6 marks)



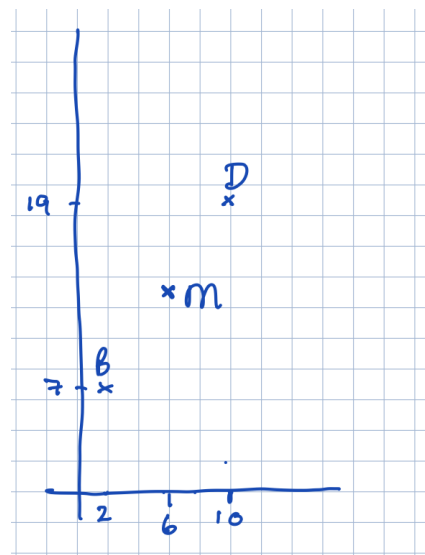
23 $ABCD$ is a kite with $AB = AD$ and $CB = CD$.

B is the point with coordinates $(10, 19)$

D is the point with coordinates $(2, 7)$

Find an equation of the line AC .

Give your answer in the form $py + qx = r$ where p, q and r are integers.



midpoint of BD

$$\frac{10+2}{2}, \frac{7+19}{2}$$

$$(6, 13)$$

gradient BD

$$= \frac{19-7}{10-2} = \frac{12}{8} = 1.5 \left(\frac{3}{2} \right)$$

$$\therefore \text{gradient of } AC = -\frac{2}{3}$$

$$y = -\frac{2}{3}x + c \quad \text{passes through } M \begin{matrix} (6, 13) \\ x \quad y \end{matrix}$$

$$13 = -\frac{2}{3} \times 6^2 + c \quad c = 17$$

$$\text{line} \Rightarrow y = -\frac{2}{3}x + 17$$

$$\times 3 \quad 3y + 2x = 51$$

(Total for Question 23 is 5 marks)



- 24 A particle P is moving along a straight line that passes through the fixed point O . The displacement, s metres, of P from O at time t seconds is given by

$$s = t^3 - 6t^2 + 5t - 4$$

Find the value of t for which the acceleration of P is 3 m/s^2

$$v = \frac{ds}{dt} = 3t^2 - 12t + 5$$

$$a = 6t - 12$$

$$\text{so } 3 = 6t - 12$$

$$6t = 15$$

$$t = \frac{15}{6} = 2.5$$

$$t = 2.5$$

(Total for Question 24 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS

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